Exploring Punctuated Evolution Through Regression Analysis

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Abstract

In punctuated evolution, the lineages that have more internal nodes show more molecular change. Hence, the root to tip paths of them increases with the internal nodes. The same pattern can also arise if the tree is impacted by the node density artifact. In this project, I will use the simulated phylogenies where the evolution would be known. I will also write Python code to simulate the birth death trees using different speciation and extinction rates through the use of Dendropy. As a minimum viable product, I will fit both linear and power-law regression models and then compare them with AIC, and then further checks the fits using the residual plots. I also plan to train the machine learning classifiers to distinguish the punctuated evolution from the gradual one as my stretch goal. Moreover, in case the evolution is positive, I would get the slope values to be greater than 0.

Background

In molecular phylogenies, evolutionary change is recorded as the branch lengths which are then measured in substitutions per site. Moreover, under the gradual evolution, the changes accumulates steadily with time. However, under punctuated evolution, a small burst of change occurs at every speciation event with tips having more nodes having the longer path lengths. Because of this very reason, Webster et al. (2003) and Pagel et al. (2006) predicts that speciation makes a larger contribution to molecular evolution.

Additionally, Venditti et al. (2006) found that the same positive relationship can also be found in some trees if they are impacted by an artifact that is called as the node density artifact. This effect happens more in trees with longer internal branches. Hence, the inference software cannot see the substitutions that occur on these branches. Due to this very reason, the paths with less nodes ends up having shorter branches whereas the paths with more nodes ends up having longer branches (Venditti et al., 2006; Hugall and Lee, 2007). In empirical datasets, it is also not possible to know which process produced this pattern. This is the reason why I will be using simulations.

Research questions/Hypotheses

1. Would path lengths increase with the increase in the internal nodes in both gradual and punctuated evolution?
   Hypothesis: the beta would be greater than zero in punctuated evolution but close to zero in gradual evolution.
2. Which is better between linear or non-linear (power-law) model to describe node count and path length relationship?
   Hypothesis: the linear model will be better fit for trees with punctuated evolution.
3. Does speciation and extinction rate along with the tree size impacts the detection rate for punctuated evolution?
   Hypothesis: Weaker detection for small trees and trees with high extinction rate, because extinct lineages leave hidden speciation events that are not visible as nodes.
4. *(Stretch goal)* Can machine-learning classifiers differentiate gradual from punctuated trees, and do they perform better than a regression-based rule?

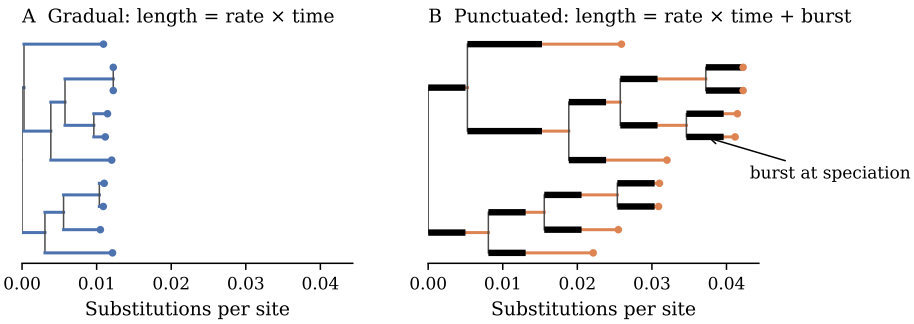


Figure 1: The simulated birth death trees with 10 tips. (A) The branch lengths only depends on rate and time in gradual evolution. (B) In punctuated evolution, every branch has some bursts of evolution. Hence, branches with extra bursts have some hidden speciation events caused by the dead lineages.

Methods

Coding will be done in Python (NumPy, SciPy, DendroPy, scikit-learn, Matplotlib).

Simulating trees (questions 1–3)

I will simulate the birth-death trees using Dendropy (Sukumaran and Holder, 2010). I will keep the extinct lineages present in the trees. I also plan to vary the extinction rate ($\mu/\lambda = 0, 0.25, 0.5, 0.75$) and tree size (50, 100 and 200 extant tips), with 100 replicate trees in each setting. Branches will be converted from chronogram to phylogram by multiplying the length with the rate (Figure 1):

- **Gradual:** length = rate × time, with rate varying among the different branches.
- **Punctuated:** length = rate × time + κ, where κ is a difference in burst that is added at every internal node. I will also use many values of κ.

I will remove the dead lineages only after I have given them some branch lengths. This is done to make sure that their speciation events are still intact in the living branches. Moreover, under the punctuated model, the speciation events still add some extra bursts.

Extracting path lengths and node counts (questions 1–3)

I will use dendropy for every surviving tips and lineages to calculate the internal nodes as well as the path lengths. I will then be saving all the results in CSV file per tree. Additionally, to make sure my code is free of errors, I will write the test cases too. For the first test case, if the clock is strict, then every tip must have the same path length along with the same number of the living tips. From my preliminary simulations, I could see that this is true. Dendropy did not flag any dead tips. Hence, I have to find out the extinct tips by the age.

Regression analysis: minimum viable product (questions 1 and 2)

For each tree, I will fit two models that will have the path lengths L_i on node count N_i for tip i:

$$\text{linear:} \quad L_i = \beta_0 + \beta_1 N_i + \varepsilon_i, \tag{1}$$

$$\text{power law:} \quad L_i = a\, N_i^b + \varepsilon_i. \tag{2}$$

Equation 1 will be fit using least squares and Equation 2 by non-linear least squares (SciPy). Because of the fact that both the equations have the same response variable, I will use Akaike

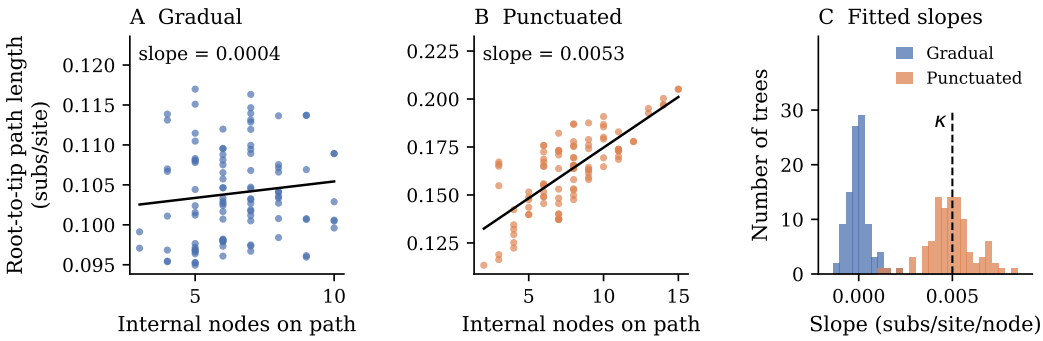


Figure 2: Initial results from my simulation settings where the tip numbers were 100 with $\lambda = 1$, $\mu = 0.5$. Here, figure A and B shows the path lengths as compared to the number of nodes both of which are fit on the linear slope. Figure (C) shows the slopes to be fit on 100 trees replicate. Moreover, the dashed line here shows the burst size.

information criterion (AIC) (Akaike, 1974). If AIC comes out to be lower, then it means that there is a better balance of fit and model complexity. I will also be making the scatter plot using the fitted lines and the residual plots to see whether each model will be fitting well or not. Moreover, the tips in the same tree share parts of their paths, so they are not independent; for this reason I will not rely on per-tip p-values, and will instead summarize the fitted slopes in 100 replicates.

Preliminary results. I have written the code for the preliminary simulations and also tried it on a 100-tip trees with ($\lambda = 1$, $\mu = 0.5$, $\kappa = 0.005$ substitutions per site). As per the expectation, the path lengths increased with the node counts in a tree that has a punctuated effect. However, the same effect was not visible in a tree that has a gradual evolution (Figure 2A, B). In all the 100 replicates, the gradual evolution slopes were coming out to be zero with (mean -0.00002), where as the tree that has the punctuated evolution has the slopes that were greater than 0 with (mean 0.0050). I also tried finding the overlap between the two types of evolutions and it came out to be looking like this with almost nothing overlapped between the two distributions(Figure 2C).

Machine learning: stretch goal (question 4)

If I have time left, I will further make a dataset that should have one row per tree with the other column having features like regression slope, R^2, the power-law exponent b (Equation 2), and the variance of path lengths. I will also be training a logistic regression and random forest classifiers by using scikit-learn (Pedregosa et al., 2011). Then I will evaluate it using the test set using confusion matrix. Further, I will compare them with a simple regression-based rule that actually classifies the trees as punctuated if the slope is coming out to be positive.

Timeline

This project of mine is expected to be done in almost seven weeks (Table 1). In the first week, I will finish the simulation code and its test cases that are mentioned in Sections 3.1–3.2. In the second week, I will run all of my simulations and then I would collect the nodes counts and the path lengths. In the third week, my aim would be to complete my minimum viable product which is fitting the linear as well as the power-law models and then compare both of these models with AIC along with their residual plots (Section 3.3). In the next two weeks, I would make and polish my final figures and then subsequently preparing the machine learning features. In week 6, I would

train my classifiers (Section 3.4) and then subsequently organize the research notebook. In my last week, I will write the final paper. If everything goes well while working on my minimum viable product, I would be done with it in the week of November 15. I will then consequently start working on my stretch goal which is a machine learning project.

Table 1: Prospective timeline

Dates	Task	Methods
Oct 26 – Nov 1	Simulation code and test cases	3.1, 3.2
Nov 2 – Nov 8	Work on simulations; collect path lengths and node counts	3.1, 3.2
Nov 9 – Nov 15	Regression analysis, AIC comparison, residual plots (MVP)	3.3
Nov 16 – Nov 29	Make and polish regression figures; make features for machine learning	3.3, 3.4
Nov 30 – Dec 6	Train classifiers; organize research notebook	3.4
Dec 7 – Dec 11	Write the paper	–

Broader impacts

A lot of studies claim that speciation drives molecular evolution because of the apparent relationship between node count and path length. By using simulations to see the relationship, the project will help me judge how reliable those claims are. This is especially important in phylogenetics where rates are used to find when species diverge. As per Douglas et al. (2025), when we ignore the bursts at speciation events, that can overestimate the divergence times. Finally, the simulations used here in this project can also show that it can test statistical method before it is applied to the real data.

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